Neural Network Compression for Ventricular Arrhythmia Detection

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Introduction

Motivation:

- Patients at high risk of Sudden Cardiac Death (SCD) rely on Implantable Cardioverter-Defibrillators (ICDs) to deliver life saving shocks during Ventricular Arrhythmias (VA).
- Current industry practices rely on simple, rule based detection algorithms
- Deep learning can improve upon ICD design by automatically extracting hidden features from Intracardiac Electrograms (IEGM) that human experts might miss.
- Standard ML models are far too memory intensive and computationally heavy to deploy on these power constrained medical edge devices.

Objective:

- Compress a Convolutional Neural Network for IEGM signal classification without significantly degrading performance (measured via F-beta scores).

Approach:

- High level architectural redesign utilizing Depthwise Separable Convolutions (DSC) along with Batch Normalization Folding.
- Quantization to lower model size by lowering precision of model parameters.

Methodology

- **Depthwise Separable Convolutions (DSC):** Replaced standard 1D convolutions with depthwise and pointwise operations, reducing the parameter count.
- **BatchNorm Folding:** Mathematically absorb Batch Normalization layers into preceding convolutional layer. This helps reduce model parameter count in a lossless manner.
- **Quantization-Aware Training (QAT):** Implemented QAT to simulate low precision forward and backward passes during fine tuning, This helps the model adapt to quantization noise.
- **Baseline Architecture:** The reference model is a 5-layer 1D CNN utilizing standard convolutions, Batch Normalization, and ReLU activations.
- **Measuring Compression:** Model footprint is calculated theoretically based on parameter count multiplied by the precision bit-width.

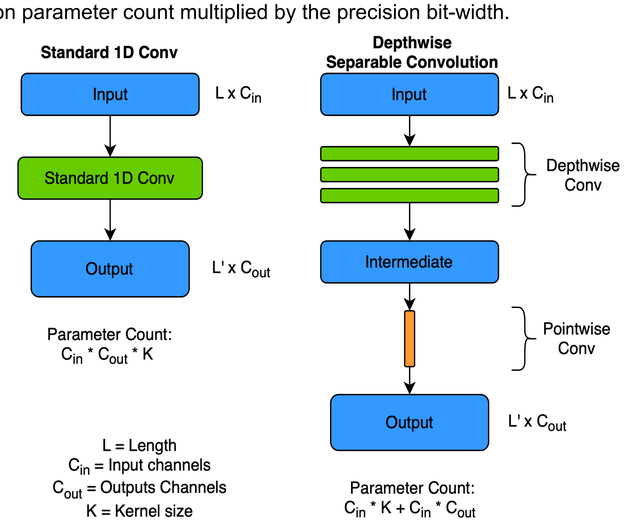

Figure 1: Architectural Comparison of Standard Convolutions and Depthwise Separable Convolutions

Results

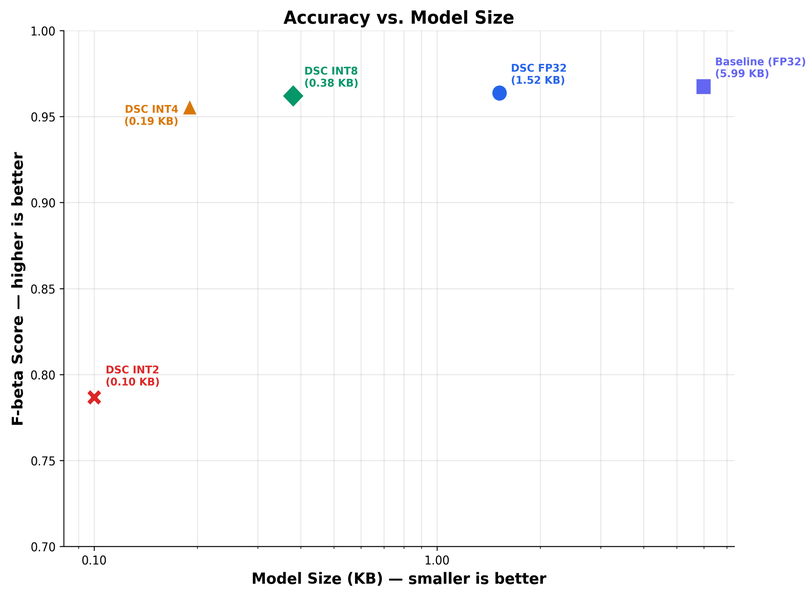

Figure 2: Model Size vs F-beta Score

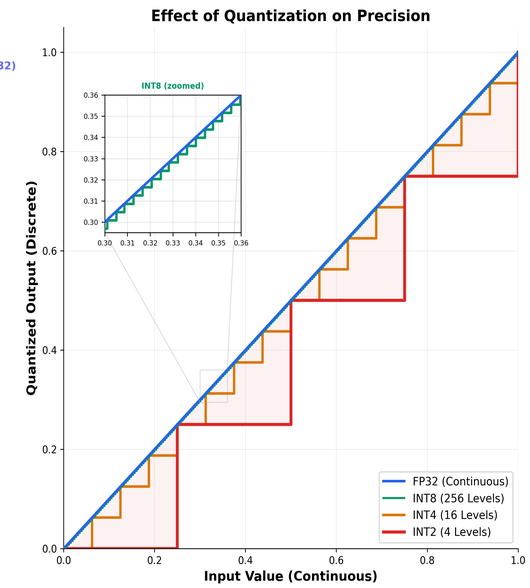

Figure 3: Visualization of Discretization From Quantization

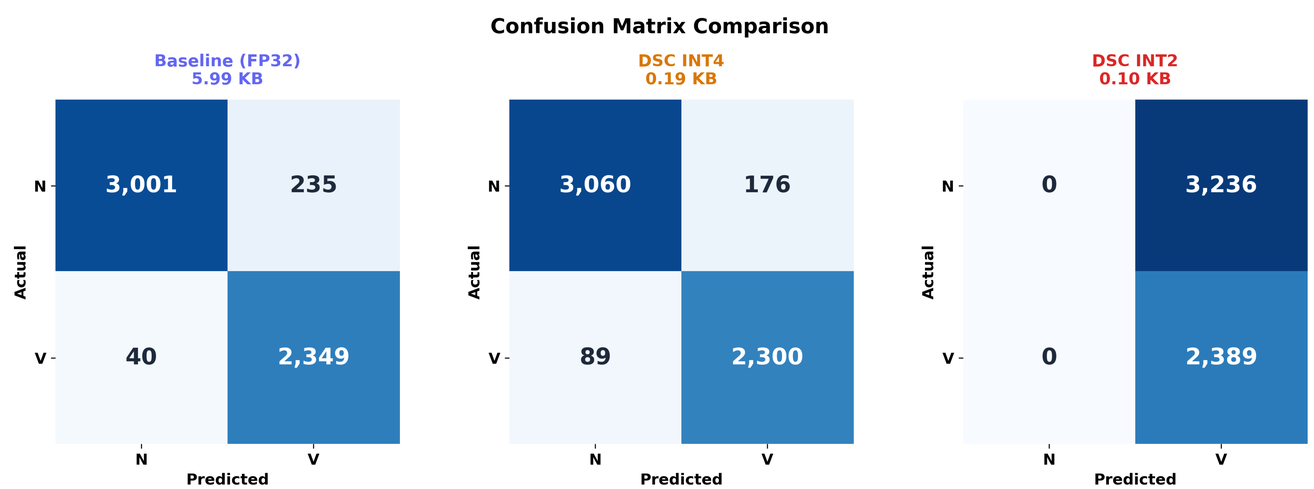

Figure 4: Confusion Matrices Across Quantization Levels

- **Model Reduction:** Replacing standard convolutions with DSCs, using BatchNorm Folding, and applying INT4 quantization reduced the model footprint by 96.8% (from 5.99 KB to 0.19 KB).
- **Metric Preservation:** The INT4 model maintained a highly competitive F-beta score of 0.956, compared to the FP32 baseline of 0.964 (See Figure 2).
- **Limit of QAT:** Quantization down to 4 discrete levels (INT2) resulted in severe precision loss. This causes the model to completely collapse, predicting only one class (See Figure 4).

Conclusion

- DSC + INT4 QAT is viable for ventricular arrhythmia classification. A 31x reduction in model size was achieved with less than 1% degradation in F-beta score. This demonstrates that aggressive quantization is practical for resource constrained devices.
- The new Depthwise Separable Convolution model reduced parameter count by 3.7x (1,534 => 422). BatchNorm folding further eliminated 32 parameters (422 => 390) with no impact on accuracy by absorbing normalization directly into convolution weights. Finally, INT4 quantization compressed remaining parameters from 32 bits to 4 bits.
- With only 4 discrete levels, the INT2 model loses the representational capacity needed for binary classification, collapsing to single class prediction. This establishes INT4 as the minimum viable bit width for this architecture.
- Future work may explore deployment of the optimized models on MCU hardware to benchmark inference latency and generate more robust results.

Acknowledgements

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Key References

[1] Z. Jia et al., "TinyML Design Contest for Life-Threatening Ventricular Arrhythmia Detection," IEEE TCAD, 2023.

[2] F. Chollet, "Xception: Deep Learning with Depthwise Separable Convolutions," CVPR, 2017.

[3] G. Franco et al., "Xilinx/brevitas," Zenodo, 2025.